

Application of YOLO Deep Learning Model for Real Time Abandoned Baggage Detection

Tossaporn Santad, Piyarat Silapasupphakornwong, Worawat Choensawat, and Kingkarn Sookhanaphibarn

*BU-Multimedia Intelligent Technology Laboratory,
School of Information Technology and Innovation
Bangkok University, Thailand*

Abstract—We proposed an abandoned-baggage detection system that the baggage was left in public places for security reasons, i.e., subway stations. The proposed system applied the YOLO deep learning model for object detection, and presented a GUI for supporting a parameter setting. With this GUI, the detection system will be invariant to lighting and camera position.

Index Terms—cctv, image detection, surveillance, yolo, forensic

I. INTRODUCTION

The idea about problem of abandoned luggage detection was established by Sugrue and Davies in around 2005 [Sugrue and Davies 2005]. Then, there are many techniques and systems were proposed to solve this problem until present. For example Kevin et al. [Kevin et al. 2006] proposed the Macov Chain Monte Carlo tracking (MCMC). Guler et al [Guler et al. 2007] proposed the Motion History Tracking. The system of alarming and finding the owner of left baggage was proposed by Chang et al.[Chang et al. 2010].

However, most of all passed research used the background subtraction method; it made many defects of the system. For example, it was not invariant with camera plane, only detect the static and specify objects and area, etc. Also most system has to adjust the plane of camera to the normal angle (eye-level) before analysis. Moreover, when the light is changed irregularly, for example, outdoor situation, most of system still cannot robust.

We applied an intelligent deep learning method for object detection, named YOLO [Redmon and Farhadi 2015], to our system that helps us to detect the baggage efficiently in the scale of real-time processing. However, YOLO can provide us a list of detected objects in each frame of video without any information about relationship between other object in the same frame and movement of the same object from previous frame.

We proposed a system that can identify the person carrying a baggage in surveillance video and tracking movement of that person and his/her baggage to analyze the suspicious activity (dropping and leaving his/her baggage). In this system, we also designed and developed a GUI for a parameter setting. With this GUI, the detection system will be invariant to lighting and camera position.

II. PROPOSED ABANDONED-BAGGAGE DETECTION SYSTEM

This study aims to detect and track an abandoned baggage in a surveillance video, presented with GUI interface. The system structure of our proposed system is shown in Fig. 1.

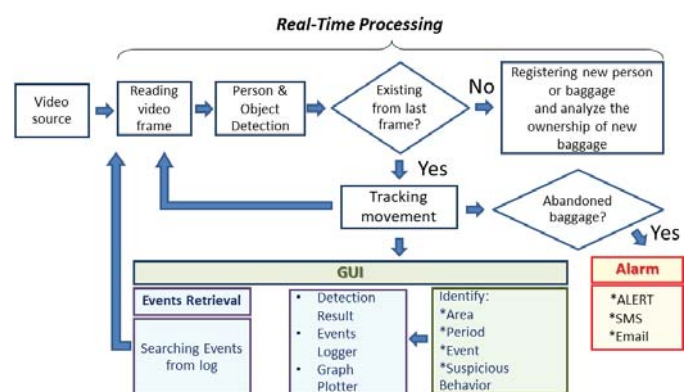


Fig. 1. A proposed system structure

The input video from surveillance camera passed through the system. Each frame of image goes through YOLO processing and give out a list of detected objects. The system will compare each object with object list from the last frame. If there are any new person or object, system will create a tracker class for them. Then system will compare each existing object with data from tracker class to analyze their movement. If any person were walking away from their baggage in a distance longer than the threshold distance set up by system administrator, the system will alarm. Finally, system can save the list events to a log file for further use as crime investigation tool.

Fig. 2 is the graphic user interface (GUI) of our system. The left top is the original video. The left bottom is video with person and bag detection. The right side from top to bottom are the console for controlling, movement history tracking, detection statistic and events log.

III. EXPERIMENTS AND RESULTS

A. Experimental setup

Experiments were conducted under 11 conditions of different lighting and camera positions and walking direction of the

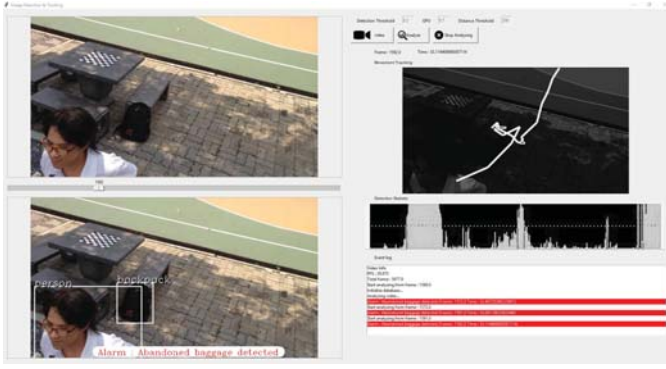


Fig. 2. Graphic User Interface of our proposed system.

subject. Using iPhone attached with a tripod as a surveillance camera. Table I presents all conditions set up of 11 cases.

TABLE I
SAMPLE CASES VARY WITH LIGHT SOURCES, CAMERA POSITION, AND
ANGLE BETWEEN WALK DIRECTION AND CAMERA PLANE

Case	Location and Light Source	Camera Position	Walk Direction
1	Outdoor	High	90
2	Outdoor	High	0
3	Outdoor	Normal	90
4	Indoor - Fluorescent Light	High	45
5	Indoor - Fluorescent Light	Normal	45
6	Indoor - Window Light	High	0
7	Indoor - Window Light	Normal	0
8	Indoor - Blue Light	High	90
9	Indoor - Blue Light	Normal	90
10	Sky Train Station	Normal	45
11	Subway Station	Normal	45

In the experiment, we varied the lighting to four conditions, which is, outdoor (sunlight), indoor with fluorescent lamp, indoor with sunlight from the window that had much difference luminance between two areas and blue light (colored fluorescent lamp). We captured the video with two camera angles, which is, high angle (from ceiling), and normal angle (eye level). The subjects, who need to carry a baggage into the surveillance area carrying a backpack. Then he will leave his backpack at the drop point and walk away from surveillance area. The walk direction was classified into 3 directions which is 0 degree (along), 45 degrees (diagonally) and 90 degrees (cross) with a camera plane. Fig. 3 illustrates the experiment set up when we captured each video. The design of the experiment allowed us to determine the parameters of our system to work under vary of places.

B. Experimental Results

Our system had achieve in detected the person and his bag and alarm when he abandoned it for all 11 cases as shown in Fig. 4.

IV. CONCLUSIONS

We proposed the system to detect an abandon baggage by applying the YOLO deep learning neural network as the object

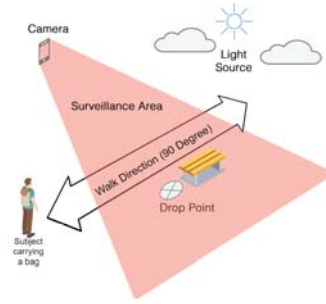


Fig. 3. An illustrate of an Experimental set up case in walking with 90 degree with sunlight and high angle of camera setting.



Fig. 4. Screenshot from our GUI showing the experimental results: (Left-side) Scenario of the person leaving his bag, (Right-side) GUI showing the alarming text of left baggage.

detection of the system. Then, the system can track movement of people and their baggage. We designed the GUI interface for alarming and retrieving pass events. The system can be successful to alarm whenever a baggage has been abandoned. The experiments showed that our system efficiently worked with invariant lighting and camera location conditions.

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